

Design & Professional Skills — First Assignment

Instructions

The following exercises are intended to get you started with Python. At the very least, you must complete the prerequisites during week 1. If you've programmed in Python before, you can probably skip the first few exercises. If you've not programmed in Python before, get as far as you can, but we don't expect you to complete everything.

This assignment is not assessed, but you do need to commit and push your solutions to your GitHub repository by Monday 12th October at 12 noon. Create separate Python files for each exercise (e.g., `exercise2.py`, `exercise3.py`, etc.) in the root directory of your repository. Your work will be automatically collected from your repository at the deadline.

GitHub Repository Instructions

You have been provided with a GitHub repository containing the necessary files for this assignment. To complete and submit your work:

1. Create separate Python files for each exercise in the root directory of your repository
2. Use clear naming conventions (e.g., `exercise2.py`, `exercise3.py`, etc.)
3. Commit your changes regularly as you work
4. Push your final solutions to your repository before the deadline
5. Your repository should contain all your Python files alongside the provided files

If you need help with Git commands, ask on Moodle or refer to the Git documentation.

Prerequisite 1

Install Python 3.11 or higher on the machine you intend to use to do your coursework. If you have problems, ask on Moodle. Alternatively, make use of study spaces in computer labs or the remote access service Desktop@UCL Anywhere.

Prerequisite 2

Install git on your machine. This will allow you to check out software and slides from github. Familiarise yourself with the basics of checking out and updating from a repository.

Prerequisite 3

Ensure you can run tkinter packages using your Python 3 installation, as we'll need these for next week's assignments. Tkinter usually comes pre-installed with python installations, but in some cases you'll need to install a separate package.

Test your install by running the `tkinter_test.py` file included in this repository.

From the command line, if python is installed as python3, you can run this as:

```
python3 tkinter_test.py
```

Exercise 1: Greatest Common Divisor Example

Type in the GCD example from `gcd_function.png` included in this repository and get it to run. We cover this algorithm in more detail in the in-person sessions and explain how it works.

Why type it in, rather than cut and paste, or checkout from github? Simply, we want you to get set up with an appropriate editor, and understand the very basics of python. If you type it in, you'll remember it. Likely you'll mistype something, and have to understand what python's error messages are telling you. And at the least, you'll need to get to grips with python's use of whitespace to indicate semantic grouping of statements. Doing this with code where you know what the answer should be is useful if you've not programmed in python before.

Exercise 2: Odd or Even

Write a program that asks the user for a number. Depending on whether the number is even or odd, print out an appropriate message to the user.

Hint: The python3 `input()` function will read input from the keyboard, but in python3 the result will be a text string, not an integer. You may need to use the `int()` function to convert it to an integer type.

Exercise 3: Fizzbuzz

Write a program that prints the numbers from 1 to 100. But for multiples of three print "Fizz" instead of the number and for the multiples of five print "Buzz". For numbers which are multiples of both three and five print "FizzBuzz".

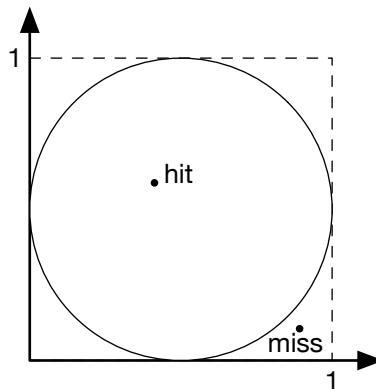
This is one of those classic interview questions that are supposed to distinguish candidates that can program from those that cannot. There are lots of versions of FizzBuzz online, but we'd like you to attempt your own before you go hunting for online solutions.

Hint: modular arithmetic might be useful.

Hint: you can do this with a `while` loop, as in the GCD example, but it's more elegant to use a `for` loop and `range` operator.

Exercise 4: Calculating Pi

We want to compute an approximation of π (3.141592...) up to a certain precision. One way to do this is by using a *Monte Carlo method* as follows. Consider the square of edge length 1 and the circle of radius $1/2$ inscribed in the square, as depicted below. Generate random points within the square. Suppose N_{hits} points hit the circle, out of a total of N_{tot} points.



Then we can approximate π by noting that N_{hits} estimates the area of the circle and N_{tot} estimates the area of the square. In fact, recalling that the area of the circle is πr^2 , for $r = 1/2$ we have

$$\frac{A_{\text{circle}}}{A_{\text{square}}} = \frac{\pi}{4} \approx \frac{N_{\text{hits}}}{N_{\text{tot}}}$$

Implement a function `estimate_pi(precision)` that returns an approximation of π up to `precision` decimal positions. Use the function `random()` from the module `random` to generate random numbers between 0 and 1. (**Hint:** the more points you generate, the more precise your approximation will be, but how do you know the precision to a fixed number of decimal places?)

Note: this is a very slow algorithm for calculating Pi, but Monte Carlo methods sometimes the only way we know that works. Not in this case though.

Exercise 5: Breaking Caesar’s Cipher

A *simple substitution cipher* is an encryption method where each letter of the alphabet is replaced by another one, according to a ciphertext alphabet. Caesar’s cipher is a very simple substitution cipher, where each letter is replaced by one n places later in the alphabet, wrapping back from Z to A if necessary. For instance, in the cipher

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	(original alphabet)
N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C	D	E	F	G	H	I	J	K	L	M	(ciphertext alphabet)

A is replaced by N, B by O and so on. Applying this cipher to the word HELLO produces the word URYYB.

Implement a function `break_cipher(text)` that takes one or more encrypted **sentences** – which is assumed to be encrypted using a Caesar cipher – finds the ciphertext alphabet used to encrypt it, and returns the decrypted string. Assume that the decrypted string will be in English, with letters A-Z and no spaces.

As there are only 26 possible Caesar ciphers, you can easily break them by hand. But how would a program automatically decide which of the 26 possible offsets is the correct one? Note that because the spaces between words in the encrypted sentences have been removed, you can’t easily search for words in the decrypted output, at least after the first word.